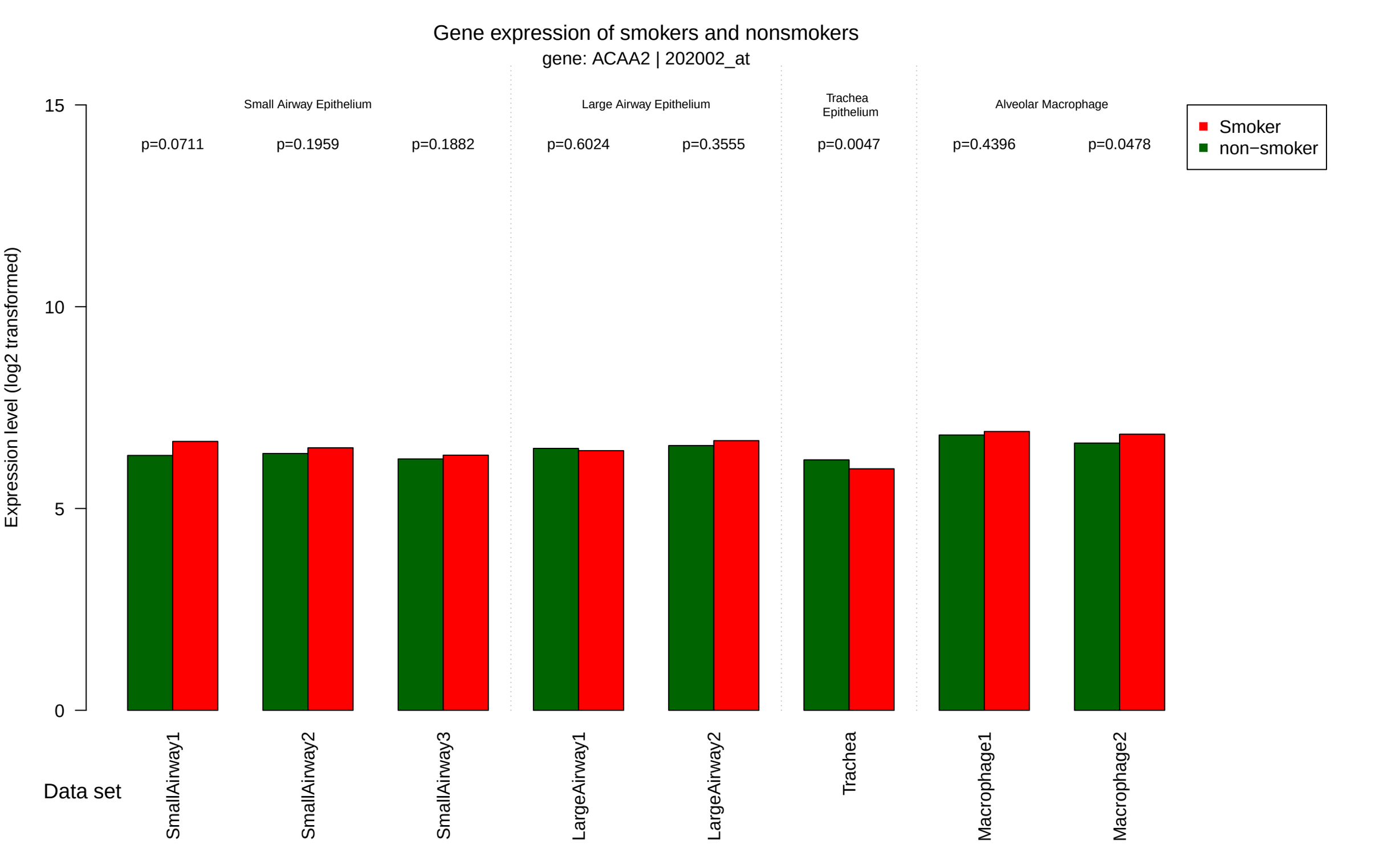
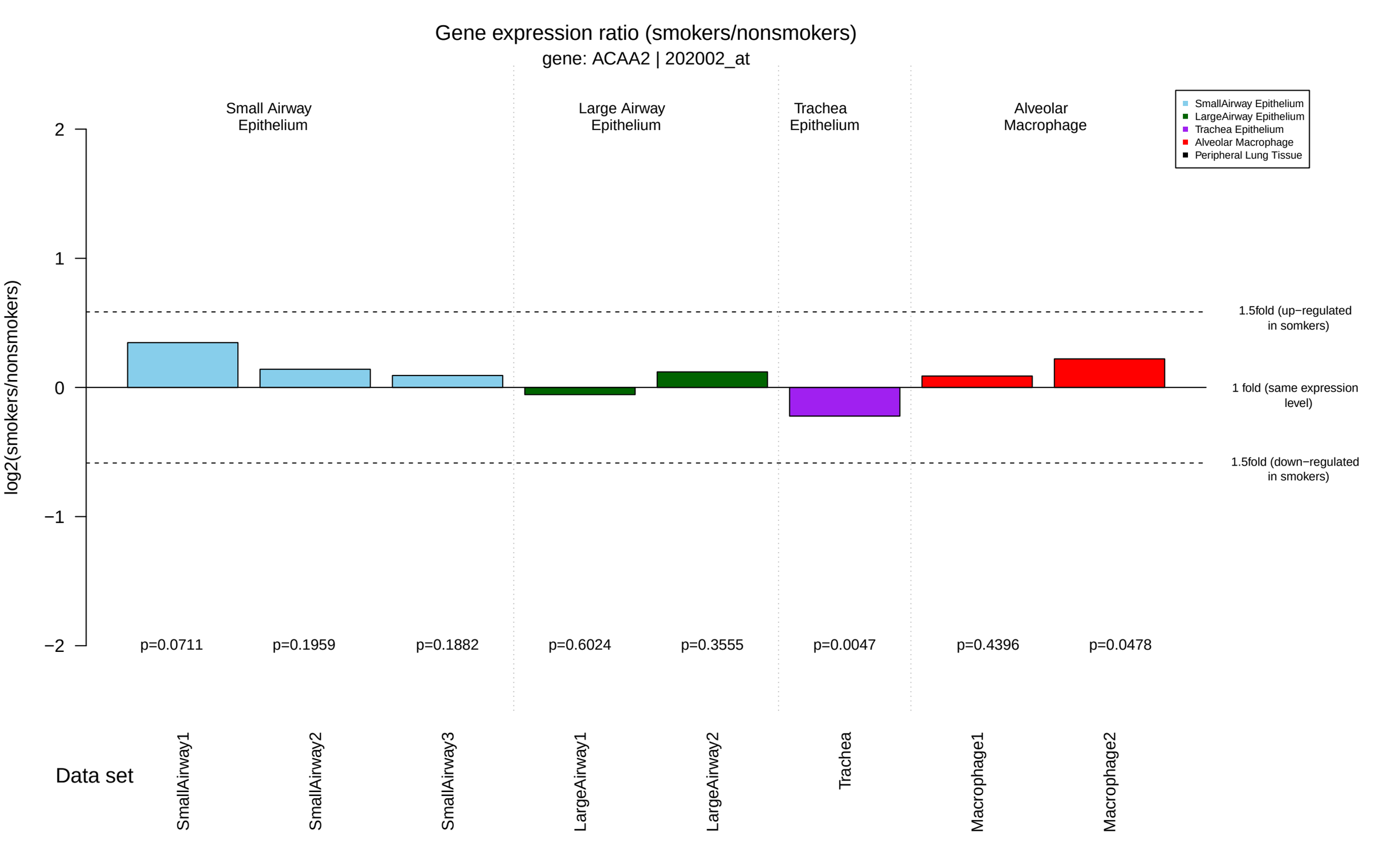
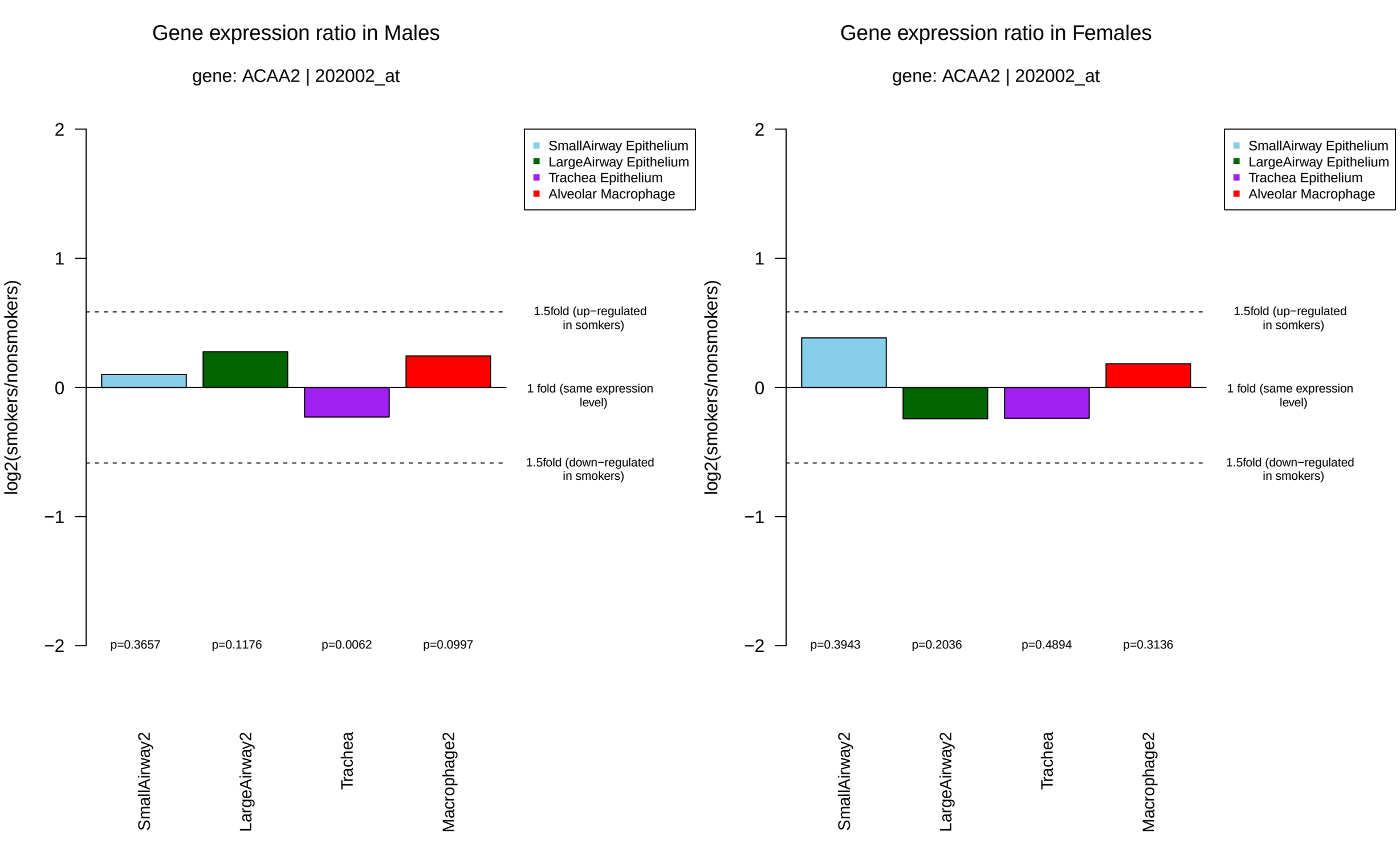
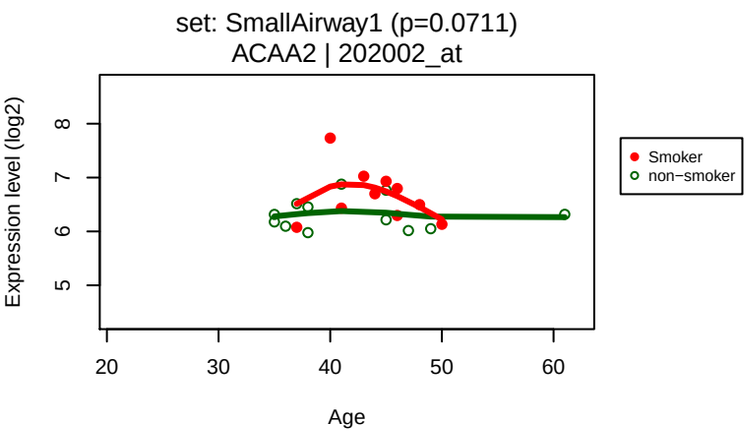
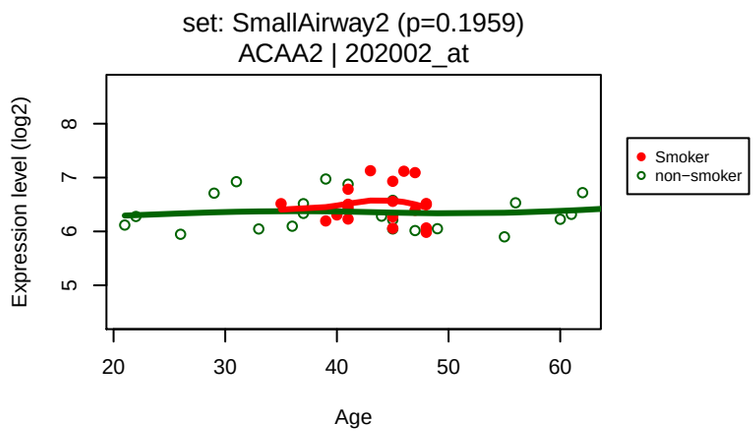
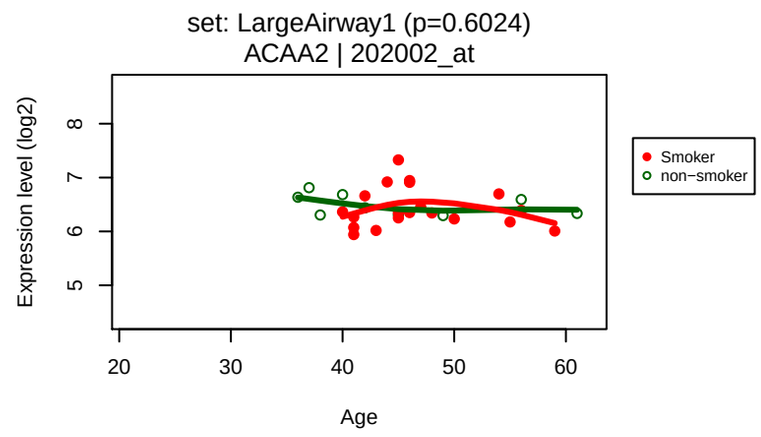

Gene expression of smokers and nonsmokers
gene: ACAA2 | 202002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0711
p=0.1959
p=0.1882
p=0.6024
p=0.3555
p=0.0047
p=0.4396
p=0.0478
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

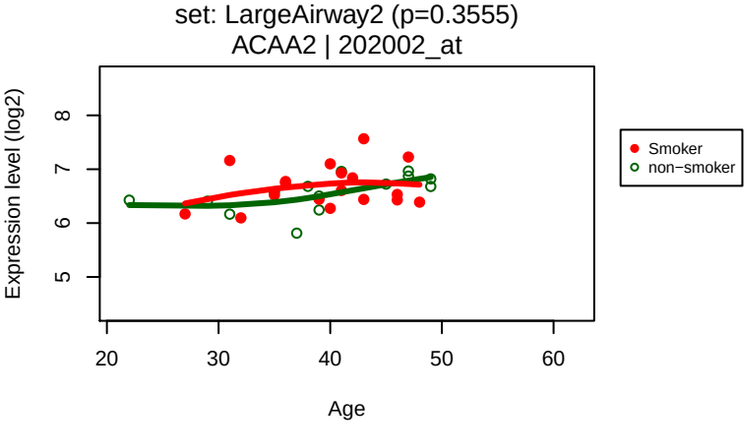

Gene expression ratio (smokers/nonsmokers)
gene: ACAA2 | 202002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0711
p=0.1959
p=0.1882
p=0.6024
p=0.3555
p=0.0047
p=0.4396
p=0.0478
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

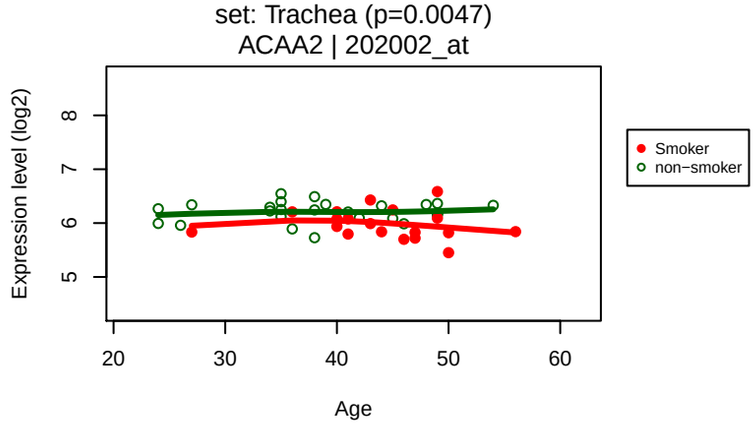

Gene expression ratio in Males
gene: ACAA2 | 202002_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3657
p=0.1176
p=0.0062
p=0.0997
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACAA2 | 202002_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3943
p=0.2036
p=0.4894
p=0.3136
SmallAirway2
LargeAirway2
Trachea
Macrophage2

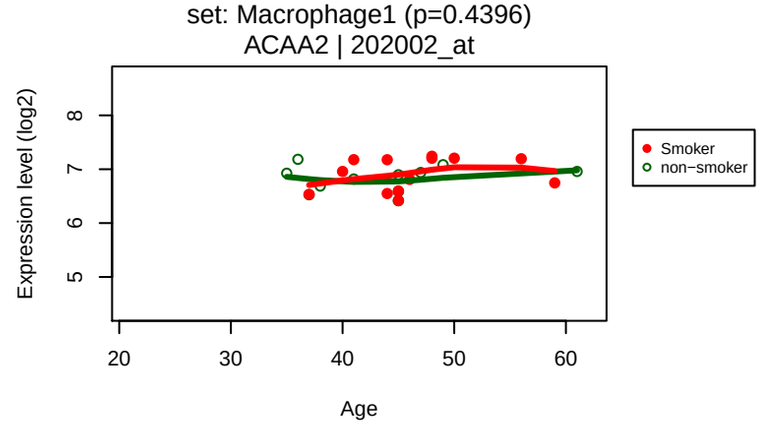

set: SmallAirway1 (p=0.0711)
ACAA2 | 202002_at
Expression level (log2)
Age
Smoker
non−smoker

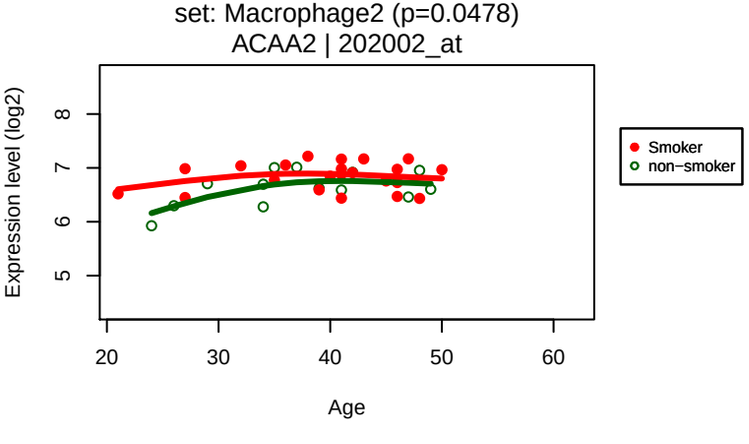

set: SmallAirway2 (p=0.1959)
ACAA2 | 202002_at
Expression level (log2)
Age
Smoker
non−smoker

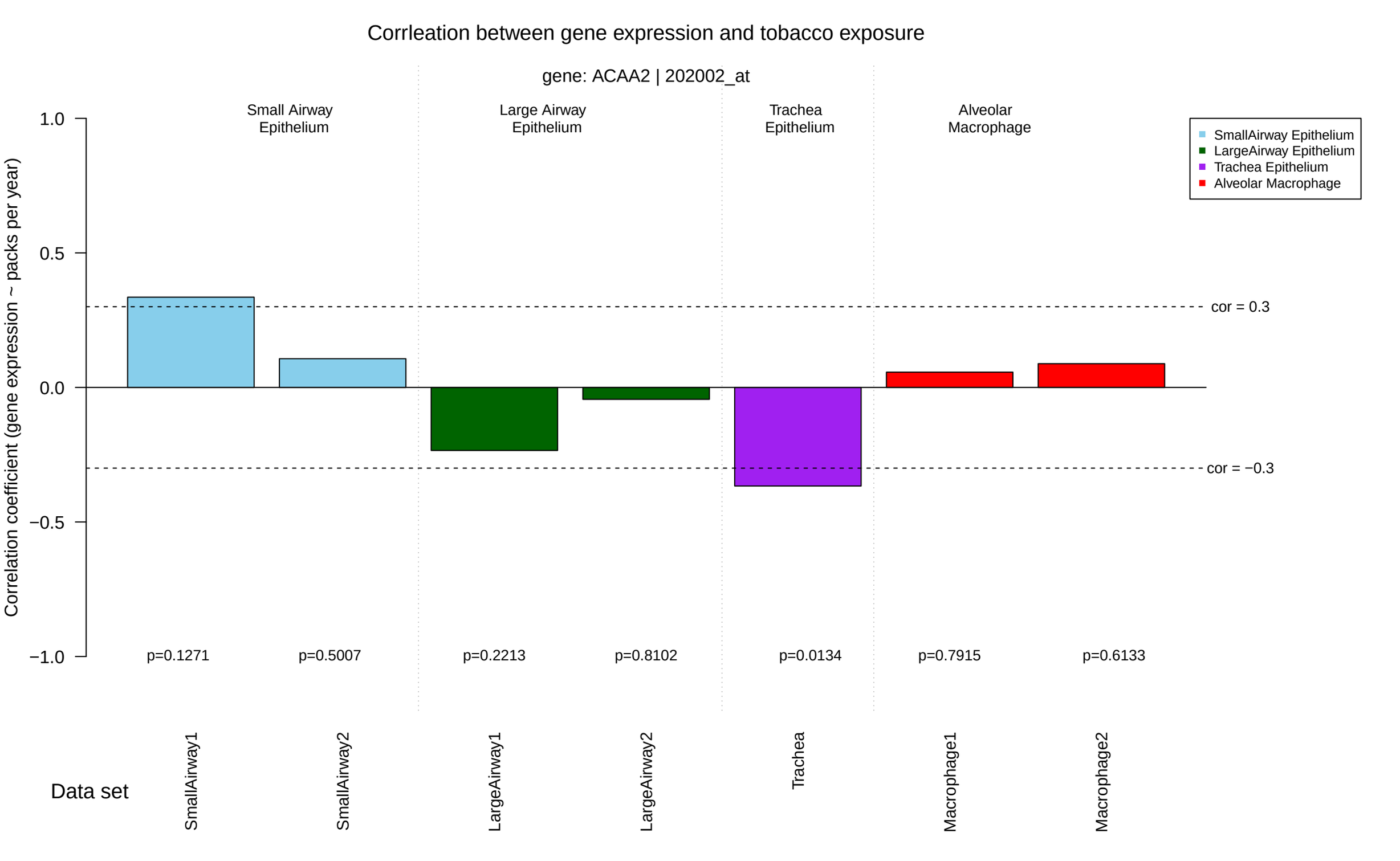

set: LargeAirway1 (p=0.6024)
ACAA2 | 202002_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3555)
ACAA2 | 202002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0047)
ACAA2 | 202002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4396)
ACAA2 | 202002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0478)
ACAA2 | 202002_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACAA2 | 202002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1271
p=0.5007
p=0.2213
p=0.8102
p=0.0134
p=0.7915
p=0.6133
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2